

proach, action interface design, context management strategy, and evaluation methodology.

3.1 Architecture and Orchestration

Despite surface-level diversity, autonomous SE agents converge on a small number of architectural patterns. As illustrated in the *Orchestration*, *Action Interface*, and *Context Management* branches of Figure 2, these patterns determine how agents perceive their environment, what actions they can take, and how they manage repository-scale context.

The observe–think–act loop. The dominant paradigm is a ReAct-style [29] loop in which the agent alternates between (i) observing the environment (file contents, test output, error traces), (ii) reasoning about what to do next (chain-of-thought or structured plan), and (iii) executing an action via a tool call. SWE-agent [27] and OpenHands [22] both instantiate this loop, differing mainly in the *action space* exposed to the model (Figure 3).

Agent–Computer Interface (ACI). Yang et al. [27] introduce the ACI concept, arguing that LLMs are a “new category of end users” that need purpose-built interfaces rather than raw shell access. SWE-agent’s ACI includes a windowed file viewer (100 lines), a linter-guarded `edit` command, summarised search results, and compressed observation history. Ablations show that each ACI component contributes measurably: removing the linter drops performance by 3 percentage points; switching to iterative search costs 6 pp; and showing full file content instead of a 100-line window costs 5.3 pp on SWE-bench Lite.

Code-as-action. OpenHands adopts the CodeAct paradigm [21], consolidating all agent actions into executable Python and bash code rather than a fixed tool API. Each turn may run arbitrary IPython cells, bash commands, or browser interactions inside a Docker-sandboxed environment. This design maximises flexibility at the cost of a larger action space that the model must navigate.

Context management. Repository-level tasks easily exceed LLM context windows. Agents employ several mitigation strategies: SWE-agent collapses older observations beyond the last 5 turns; AutoCodeRover [30] parses the codebase into an AST and exposes structure-aware search APIs

(`search_class`, `search_method_in_class`) that return only signatures, not full definitions; Agentless [25] constructs a hierarchical repository skeleton and uses embedding-based retrieval to select candidate files before prompting the LLM.

3.2 Representative Systems

We survey five prominent systems that span the design space. Table 1 provides a comparative overview.

SWE-agent [27]. Developed at Princeton, SWE-agent wraps a GPT-4 Turbo backbone in a custom ACI with specialised file navigation, editing, and search commands. It achieved 12.47% on the full SWE-bench test set (2,294 tasks) and 18.00% on SWE-bench Lite when first published. The agent follows a typical trajectory: reproduce the issue, localise the faulty code via `find_file/search_dir`, then enter an edit–execute loop. The median successful run completes in 12 steps at \$1.21 API cost.

Agentless [25]. From UIUC, Agentless challenges the need for agentic autonomy. It replaces the interactive loop with a fixed three-phase pipeline: *localise* → *repair* → *validate*. Localisation uses a hierarchical narrowing strategy (files → classes/functions → edit locations) combining LLM prompting with embedding retrieval. Repair generates up to 40 candidate patches in a search/replace diff format. Validation selects the best patch via regression testing, LLM-generated reproduction tests, and majority voting over AST-normalised candidates. Agentless resolved 32.00% of SWE-bench Lite at an average cost of only \$0.70 per problem, making it the top open-source system at the time and establishing a strong non-agentic baseline. OpenAI later adopted Agentless as its default scaffold for showcasing GPT-4o and o1 on SWE-bench Verified.

AutoCodeRover [30]. From NUS, AutoCodeRover takes a “software-engineering-oriented” approach, operating on program representations rather than raw files. It parses the codebase into an AST and provides the LLM with structure-aware search APIs. A stratified context retrieval process iteratively expands the agent’s understanding, starting from issue keywords and refining through multiple search rounds until sufficient context is gathered. When a test suite is available, Spectrum-Based Fault Localisation (SBFL)

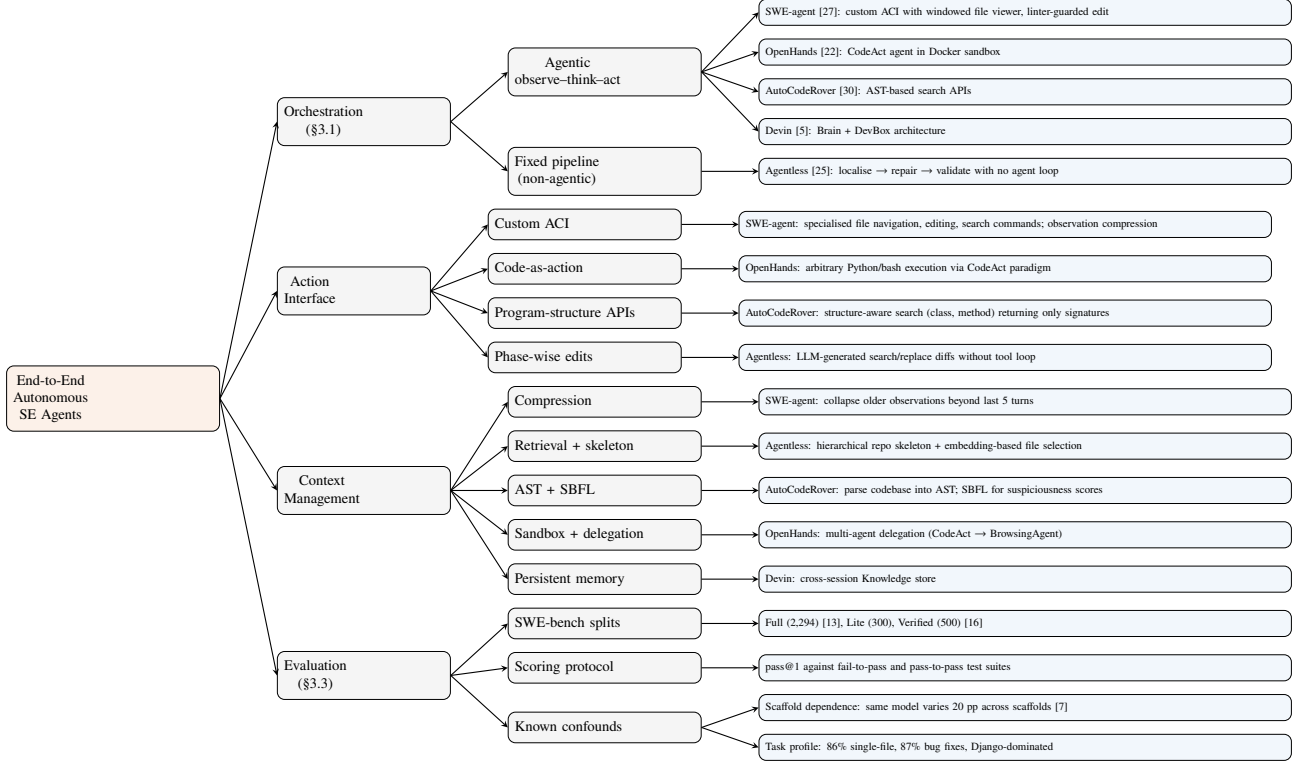


Figure 2 Taxonomy of end-to-end autonomous SE agents. Systems are classified along orchestration approach (agentic loop vs. fixed pipeline), action interface design (custom ACI, code-as-action, structure-aware APIs), context management strategy (compression, retrieval, AST, delegation, memory), and evaluation methodology (benchmark splits, scoring, known confounds). Representative systems are shown as leaf exemplars.

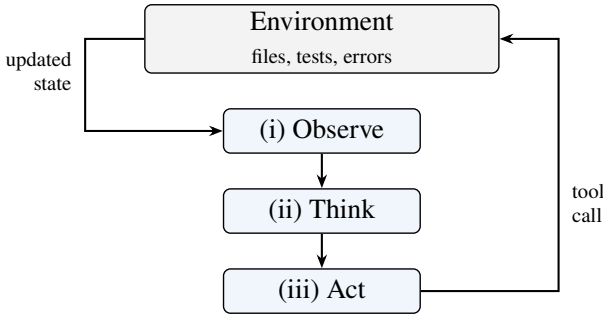


Figure 3 ReAct-style observe-think-act loop. The agent reads environment state, reasons about the next step, executes a tool call, and repeats.

augments the search with suspiciousness scores. AutoCodeRover achieved 19% on SWE-bench Lite in a single run and 26% with three retries, at a cost of \$0.43–\$1.30 per task.

OpenHands [22]. Formerly OpenDevin, OpenHands is an open-source platform (ICLR 2025, 32K GitHub stars, 188 contributors) built around the CodeAct agent. Each session runs inside a Docker container with a bash shell, IPython server, and Chromium browser. The platform supports

multi-agent delegation—a generalist CodeAct agent can offload web browsing to a specialised BrowsingAgent. On SWE-bench Lite, CodeAct v1.8 with Claude 3.5 Sonnet achieved 26.0%. On SWE-bench Verified, a later submission reaches 60.6% with a single rollout and 66.4% when five trajectories are reranked by a dedicated critic model [17].

Devin (Cognition) [5]. Devin is a proprietary cloud-hosted agent positioned as an “autonomous AI software engineer.” Its architecture separates a *Brain* (reasoning layer) from a *DevBox* (execution VM with shell, IDE, and browser). A persistent Knowledge store provides cross-session memory. Cognition reports strong full-benchmark results, but independent leaderboard analysis finds 68.3% on SWE-bench versus 48.2% on SWE-bench Verified (a 29% relative drop), suggesting possible overfitting to easier unfiltered instances [6].

Taken together, these five systems expose a recurring trade-off between agentic flexibility and pipeline efficiency. Agentless shows that a fixed localise–repair–validate pipeline can outperform interactive agents on SWE-bench Lite at lower cost (\$0.70 vs. \$1.21), chal-

Table 1 Comparison of end-to-end autonomous SE agents. Architectural patterns: **OTA** = observe–think–act loop, **ACI** = custom Agent–Computer Interface, **CaA** = code-as-action, **Pipe** = fixed multi-phase pipeline (no agent loop). SWE-bench scores are single-pass (pass@1) unless noted; **Lite** = SWE-bench Lite (300 instances), **Full** = SWE-bench Full (2,294 instances). Cost is average USD per task instance.

System	Origin	Architecture	Context Strategy	Lite	Full	Cost
SWE-agent	Princeton	OTA + ACI	Observation compression	18.0%	12.5%	\$1.21
Agentless	UIUC	Pipe	Hierarchical skeleton + embeddings	32.0%	–	\$0.70
AutoCodeRover	NUS	OTA + AST search	AST-based APIs + SBFL	19% [†]	12.4%	\$0.43
OpenHands	Multi-inst.	OTA + CaA	Docker sandbox + delegation	26.0%	–	–
Devin	Cognition	OTA + Brain/DevBox	Persistent knowledge store	– [‡]	–	propr.

[†]26% with 3 retries (pass@3). [‡]Proprietary; see text for Verified estimates.

Table 2 SWE-bench dataset splits.

Split	Size	Selection criterion
Full	2,294	All filtered PRs
Lite	300	Subset selected for tractability
Verified	500	Human-validated

lenging the assumption that autonomy is necessary for strong results. Among agentic designs, interface and context strategy matter as much as the backbone LLM: SWE-agent’s ACI ablations, OpenHands’s CodeAct action space, and AutoCodeRover’s AST search APIs each demonstrate that how an agent reads and edits code can shift resolve rates by several percentage points at comparable model capacity.

3.3 Full-Pipeline Evaluation

The *Evaluation* branch of Figure 2 summarizes the primary benchmark splits, scoring protocols, and known confounds that shape current performance reports.

SWE-bench: dataset construction. The primary evaluation instrument for end-to-end agents is SWE-bench [13]. It was constructed by scraping merged pull requests from 12 popular open-source Python repositories (including Django, scikit-learn, sympy, matplotlib, and Flask). Each PR was filtered to retain only those that modify the project’s test suite. For each retained instance the benchmark provides: (i) the repository snapshot *before* the PR was merged, (ii) the issue description (title + body), (iii) a gold patch (the human-written fix), and (iv) a set of *fail-to-pass* tests (which fail before the fix and should pass after) and *pass-to-pass* tests (which must continue to pass). An agent receives only items (i) and (ii); its generated patch is applied and scored against both test sets.

Splits and filtering: **Full** comprises all filtered instances. **Lite** selects 300 instances for faster iteration, but Xia et al. [25] found quality issues—including gold-patch leakage in issue text and tasks that lack sufficient information to be solvable. **Verified** [16] addresses these concerns: human annotators assessed each Full instance for problem clarity, test correctness, and solvability, retaining 500 high-quality instances that have become the de facto leaderboard standard [7].

Task complexity profile. Although SWE-bench tests real-world issues, most tasks are relatively small [8, 7]. In the Verified split, human annotators estimated that 39% of tasks are trivial (<15 min fix), 52% are small (15 min–1 h), 8% are substantial (1–4 h), and only 3 tasks were rated >4 h. These estimates correlate with patch size: easy tasks average 5 changed lines of code (LOC) and 1.0 files; medium tasks average 14 LOC and 1.3 files; hard tasks average 56 LOC and 2.0 files. Crucially, 86% of instances are single-file changes, and 87% are classified as bug fixes rather than feature requests or refactorings. The dominance of Django (nearly 50% of instances) and four other repositories (>80% combined) further limits diversity.

Leaderboard trajectory. Figure 4 charts the rapid progress on SWE-bench from late 2023 through mid-2026. In October 2023, the best RAG baselines resolved 1.96% of Full and 0.33% of Lite [13]. SWE-agent’s agentic loop lifted Full to 12.5% and Lite to 18.0% in early 2024. On the Full set, AutoCodeRover reached 18.0% (pass@3) and Devin 13.9% around the same period. The scaffolding race then accelerated both tracks: by late 2024, Agentless 1.5 with Claude 3.5 Sonnet reached 40.7% on Lite; by mid-2025 the best Lite score stood at 60.3%. On the Full 2,294-instance set, fewer teams report results post-2024, but dataku’s independent analysis [6] records

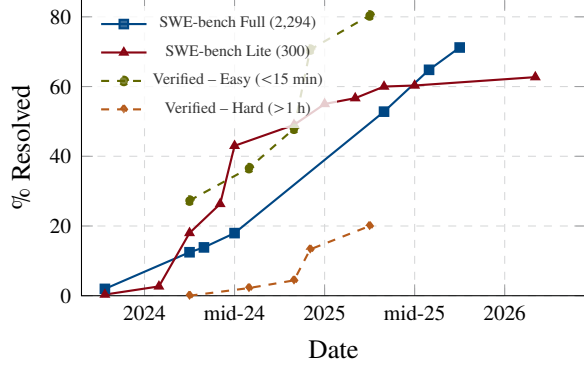


Figure 4 SWE-bench progress over time. Solid lines: best-in-class resolve rate on the Full set (blue, 2,294 tasks) and Lite (red, 300 tasks). Dashed lines: performance on easy (<15 min, green) vs. hard (>1 h, orange) subsets of Verified, illustrating the widening difficulty gap [8]. Data compiled from leaderboard submissions [20, 6, 1].

Claude Opus 4 at 71.2% and Devin at 68.3%. On Lite, the frontier reached ~63% by early 2026 [20].

The easy-hard gap is striking: the best individual system solves ~80% of easy tasks (<15 min) but only ~20% of hard ones (>1 h) [8]. Combined across all top systems, 95% of easy instances have been resolved by at least one agent, versus only 42% of hard ones—and among hard multi-file issues, only 9 of 25 have ever been solved. This ceiling effect, combined with contamination risk (half the issues predate 2020 and the repositories are widely used as training data [7]), raises concerns about the benchmark’s continued discriminative power. SWE-bench Pro, a harder leaderboard with novel uncontaminated problems, shows Claude Opus 4.5 dropping to ~46% [1].

Beyond SWE-bench. SWE-bench covers only Python bug-fixing in 12 repositories. Newer benchmarks begin to address broader scopes: SWE-bench Multilingual (300 tasks, 9 languages), CodeAgent-Bench (interactive development environments), and real-world deployment metrics. A comprehensive discussion of the evaluation landscape is deferred to §7.2.

4 Planning and Search Strategies for Code Synthesis

Planning is the stage in which an agent turns a vague intent into an operational hypothesis about *what* code should change and *how* the change can be produced. In the vibe coding pipeline, this stage sits between intent capture and code generation, but in practice it is rarely a single up-front step. Agents plan while

Table 3 Planning and search mechanisms in LLM-based code synthesis.

Mechanism	Planning object	Role in code tasks
Chain / tree / graph of thoughts	Intermediate reasoning steps	Explore alternative designs, algorithms, or debugging hypotheses before emitting code
Hierarchical decomposition	Subtasks, files, functions, edit locations	Reduce repository-scale tasks into bounded local edits
Candidate sampling	Multiple complete or partial programs	Trade inference cost for higher pass@k and diversity
Execution-guided selection	Tests, generated tests, compiler output	Rank or prune candidates using external evidence rather than model confidence alone
ReAct-style re-planning	Updated belief state after tool observations	Interleave search, reading, editing, and feedback in a single loop

reading issue reports, searching repositories, choosing files, decomposing edits, sampling candidate patches, and revising their strategy after failed tests. Thus, planning is best understood as a control layer over generation rather than as a separate document written before generation begins.

This section surveys three aspects of that control layer:

1. Plan representations: determine what intermediate state the agent exposes to itself: natural-language rationales, hierarchical task lists, program sketches, or repository-localisation hypotheses;
2. Search strategies: determine how many possible solutions are explored and how candidates are ranked;
3. The plan-generate interface: determines whether planning happens once, repeatedly, or only when execution feedback invalidates the current hypothesis.

4.1 Plan Representations and Decomposition

Plan representations determine how an agent organises reasoning before generating code. The simplest is chain of thought [24]: a linear sequence that commits early to one trajectory. If the first design choice is wrong, subsequent steps elaborate the mistake. Tree of Thoughts [28] and Graph of Thoughts [2] relax this by allowing exploration, backtracking, and merging across alternative paths (Figure 5). For code synthesis, this means agents can consider multiple decomposition

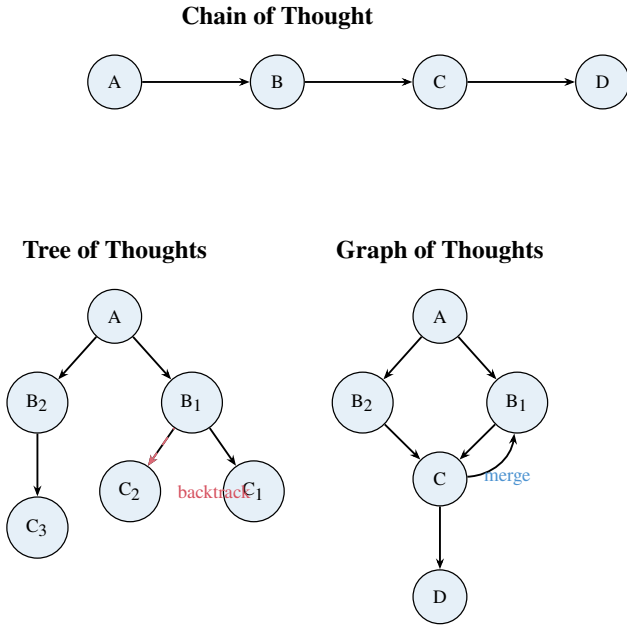


Figure 5 Plan representation structures. Chain of Thought commits to a single linear path. Tree of Thoughts explores alternatives and backtracks. Graph of Thoughts merges branches and revisits earlier states.

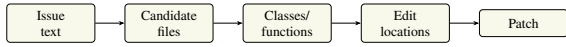


Figure 6 Hierarchical localisation pipeline. Agents narrow scope iteratively from repository-level to line-level edits.

strategies—“change the parser”, “change the caller”, or “adjust the compatibility layer”—as explicit search states rather than implicit in a single sample.

Repository-level agents usually add a more concrete representation: *localisation plans*. Agentless decomposes issue resolution into files, classes/functions, and edit locations before generating a patch [25]. AutoCodeRover uses AST-backed APIs such as `search_class` and `search_method_in_class` to plan over program structure rather than raw text [30]. These grounded representations name concrete program elements that can be validated against the repository (Figure 6), turning a broad intent into a bounded search problem.

Plans may also be expressed as *specification sketches*: partial interfaces, pseudocode, assertions, or examples that constrain the generated implementation. Useful plans are grounded in observable artefacts—issue text, existing code, API conventions, and failing tests—rather than invented requirements.

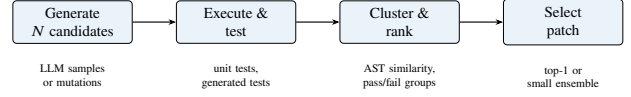


Figure 7 Candidate-selection pipeline. Systems like AlphaCode and CodeT generate many candidates, execute them against tests (real or generated), cluster by behaviour, and select a small high-quality set.

4.2 Search and Exploration Strategies

Once a representation exists, an agent must decide how broadly to search. Single-sample generation is fast, but it treats the model’s first completion as the solution. Search-based systems instead spend additional inference or execution budget to explore alternatives. The simplest form is self-consistency: sample multiple reasoning paths and select the most common or most consistent answer [23]. For programming, the same idea becomes sample-and-rank: generate several candidate programs or patches, then choose using tests, static checks, model judges, or majority voting. AlphaCode [14] and CodeT [3] exemplify this pattern (Figure 7). AlphaCode generated millions of candidates for competitive programming, filtered and clustered by execution results, then submitted a small diverse set. CodeT generates both programs and tests, selecting code consistent with generated test cases. The key insight is that LLMs often contain the right solution somewhere in the sample distribution even when top-ranked output is wrong. The limitation is self-referential objectives: if generated tests are weak, selection optimises for the model’s own misunderstanding. Execution-guided search is strongest when feedback includes external oracles—existing unit tests, compiler errors, type checkers, or repository-specific regression suites.

Search also appears in repository navigation. Agentless samples multiple locations and candidate patches, then normalises patches by AST and uses validation to pick among them [25]. AutoCodeRover expands context iteratively through program-structure queries, optionally using Spectrum-Based Fault Localisation (SBFL) to prioritise suspicious code when tests are available [30]. In both cases, search is not over complete programs alone; it is also over *where to look* and *what evidence to include*. This matters because code generation quality is bounded by context quality. A correct patch is unlikely if the agent never places the relevant function, invariant, or failing trace in context. The cost of search is the main trade-off. More samples increase diversity and pass@k, but they also increase

Table 4 Plan–generate coupling patterns.

Pattern	Mechanism	Trade-off
Plan-then-generate	Plan once, then generate code conditioned on it	Easy to inspect; brittle if plan is wrong
Interleaved planning	Alternate plan updates and tool use in observe–think–act loop	Adaptive; plan distributed across turns
Generate-then-refine	Generate, critique via feedback, revise iteratively	Self-correcting; may reopen planning after execution

latency, token cost, and validation cost. Moreover, search can overfit to narrow tests: a candidate that passes the available suite may still violate unstated requirements. Strong systems therefore combine multiple pruning signals: syntactic validity, static analysis, failing-test repair, pass-to-pass regression preservation, and semantic similarity to the intended behaviour. Search is most useful when the ranking signal is at least as reliable as the generator is diverse.

4.3 Plan–Generate Interplay

Planning and generation can be coupled in three broad ways (Table 4). *Plan-then-generate* is easy to inspect and aligns with human practice, but brittle if the initial plan is wrong [25]. *Interleaved planning* adapts to new information but distributes the plan across many turns, making it harder to audit [29]. *Generate-then-refine* treats the first output as a hypothesis and revises via self-critique or execution feedback [15], connecting planning to the refinement loop in §6.

The central design question is not whether agents should plan, but how plans remain *grounded*. Grounding mechanisms include retrieval over repository structure, AST-level symbol search, execution traces, failing tests, and explicit constraints extracted from the issue report. Effective systems keep plans close to these artefacts and treat generation as one move in a larger search process, reframing code synthesis from “write the answer” to “maintain and test a hypothesis about the required program change”.

5 Multi-Agent Frameworks for Code Generation

An alternative architectural approach to code generation distributes the work across multiple specialised agents, each responsible for a distinct phase or concern. Where single-agent systems integrate planning, generation, testing, and refinement into one LLM conversation, multi-agent frameworks decompose the software lifecycle into roles—architect, programmer, tester, reviewer—and orchestrate collaboration among agents playing those roles. This division-of-labour mirrors human software teams and aims to improve code quality, modularity, and error detection beyond what monolithic agents achieve.

This section surveys role decomposition and communication patterns (§5.1), examines representative framework designs (§5.2), and assesses when multi-agent designs outperform monolithic alternatives (§5.3).

5.1 Role Decomposition and Communication

Multi-agent systems partition code-generation tasks along two primary dimensions: *temporal stages* (design, implementation, testing, review) and *functional concerns* (requirements analysis, algorithm design, execution, debugging). Agents are assigned roles corresponding to these partitions, and collaboration protocols define how information flows among them.

Role taxonomy. Common roles mirror human software teams. A *product manager* or *architect* agent refines high-level requirements into technical specifications. A *programmer* or *coder* agent generates implementation. A *test designer* agent writes test cases independently of the code. A *test executor* agent runs code against tests and reports failures. A *reviewer* agent critiques code for correctness, style, or security. More specialised roles include a *retrieval agent* that searches documentation or past solutions, and a *debugger agent* that localises and repairs faults. Not all frameworks use all roles; designs range from three agents to dozens.

Communication patterns. Agents exchange information through structured protocols. The simplest is *sequential handoff*: the architect produces a design document, passes it to the programmer, who generates

code, passes it to the tester, and so on. More sophisticated patterns include *iterative feedback loops*, where the test executor returns failure traces to the programmer for revision, and *debate or voting*, where multiple agents propose alternatives and a consensus mechanism selects the best. Communication may use natural language (agents converse as humans would), structured artifacts (JSON schemas, API specifications, test suites), or hybrid approaches.

Memory and state management. Multi-agent systems maintain shared state through blackboard architectures, message queues, or centralised repositories. A blackboard allows all agents to read and write a common workspace containing the current design, code, tests, and execution logs. Message queues enforce ordering and ensure agents see consistent snapshots of the task. Centralised repositories (e.g., a simulated Git workflow) provide versioning and rollback. The choice of state management affects coordination overhead: blackboards are flexible but require synchronisation; message queues are deterministic but rigid; repositories add version-control semantics but increase complexity.

5.2 Framework Designs

We survey four representative multi-agent frameworks that span diverse design choices in orchestration, role granularity, and evaluation focus. Table 5 provides a comparative overview.

ChatDev [18]. From Tsinghua University and collaborators, ChatDev models a virtual software company in which agents assume executive and technical roles: CEO, Chief Product Officer (CPO), Chief Technology Officer (CTO), Programmer, Reviewer, Tester, and Art Designer. The system organises work into a waterfall-like sequence of “chat chains”: each phase (designing, coding, testing, documenting) corresponds to a multi-turn conversation among relevant agents. For example, the CPO and CEO refine requirements, then the CTO and Programmer generate code, then the Programmer and Tester validate it. ChatDev introduces *communicative dehallucination*: agents maintain memory of earlier discussions and replay relevant context to reduce inconsistencies and hallucinations across phases. The framework also supports role-playing prompts that instruct each LLM instantiation to adopt a persona, though all agents share the same base model. ChatDev demonstrates end-to-end soft-

ware generation from natural-language descriptions to executable programs, but performance metrics on standardised benchmarks are limited in the original publication.

MetaGPT [9]. MetaGPT, from DeepWisdom and collaborators, encodes human-like Standardized Operating Procedures (SOPs) into LLM prompts to streamline multi-agent collaboration. It assigns four primary roles: Product Manager, Architect, Engineer, and QA. Each role produces structured artifacts—Product Requirement Documents (PRD), system designs, implementation code, and test plans—passed sequentially along an “assembly line”. The key innovation is that agents communicate via formalised documents rather than free-form chat, reducing ambiguity and cascading hallucinations. MetaGPT includes an *executable feedback module*: generated code is run against tests, and failures trigger targeted revisions by the Engineer agent. On HumanEval and MBPP, MetaGPT achieves 85.9% and 87.7% pass@1 respectively, outperforming many single-agent baselines and prior multi-agent systems. The framework demonstrates that structured inter-agent protocols can significantly reduce errors compared to informal chat-based collaboration.

AgentCoder [10]. AgentCoder, from the University of Hong Kong and collaborators, simplifies role structure to three agents: Programmer, Test Designer, and Test Executor. The Programmer generates code. The Test Designer independently generates test cases, ensuring tests are not biased by the code. The Test Executor runs code against tests and reports failures. If tests fail, the executor provides feedback (error messages, stack traces) to the Programmer, which revises the code. This cycle iterates until all tests pass or a budget is exhausted. AgentCoder’s strength lies in *independent test generation*: because tests are written without seeing the code, they better reflect specification requirements rather than implementation quirks. On HumanEval and MBPP, AgentCoder with GPT-3.5 reaches 77.4% and 89.1% pass@1, surpassing single-agent GPT-3.5 baselines (69.5%, 63.0%) and demonstrating that even a minimal three-agent decomposition improves reliability.

MapCoder [11]. MapCoder, from Bangladesh University of Engineering and UCLA, targets competitive programming by mimicking the human problem-solving cycle through four agents: Recall, Plan, Code, and Debug. The Recall agent retrieves similar past

Table 5 Comparison of multi-agent code-generation frameworks.

System	Roles	Communication	Key mechanism
ChatDev	CEO, CPO, CTO, Programmer, Reviewer, Tester	Sequential chat-based handoffs with communicative dehallucination	Waterfall-inspired phases; memory replay to prevent hallucinations
MetaGPT	Product manager, Architect, Engineer, QA	SOP-encoded prompts with structured artifacts (PRD, design docs)	Assembly-line paradigm; executable feedback module
AgentCoder	Programmer, Test designer, Test executor	Iterative loop: code → tests → execution → feedback	Independent test generation; execution-driven refinement
MapCoder	Retrieval, Planner, Coder, Debugger	Sequential pipeline with recalling → planning → coding → debugging	Mimics human problem-solving cycle; example retrieval

problems or solution templates from a knowledge base. The Plan agent decomposes the problem into algorithmic steps. The Code agent generates the implementation. The Debug agent runs the code, identifies failures, and proposes fixes. Agents operate sequentially, passing intermediate results forward, and backtracking occurs when debugging reveals a flaw in earlier stages. MapCoder achieves strong results on competitive benchmarks: 93.9% pass@1 on HumanEval, 83.1% on MBPP, and 28.5% on CodeContests. The framework shows that task-specific role design (e.g., retrieval for competitive programming) can yield state-of-the-art performance when aligned with domain workflows.

5.3 When Does Multi-Agency Help?

Multi-agent designs introduce coordination overhead, increased token costs, and additional failure modes. Whether they outperform monolithic agents depends on task complexity, the quality of role decomposition, and the effectiveness of inter-agent communication.

Benefits: specialisation and modularity. The primary advantage of multi-agent systems is *specialisation*. When a single agent must simultaneously plan, code, test, and debug, its attention is divided and errors in one phase can cascade unchecked. Decomposing these concerns into separate agents allows each to focus on a narrower task. For example, AgentCoder’s independent test designer generates tests that are more comprehensive and less biased than tests written by the same agent that wrote the code. Similarly, MetaGPT’s structured artifacts (PRDs, design documents) force explicit planning before coding, reducing the likelihood that implementation drifts from requirements. Modularity also enables targeted improvement: if test generation is weak, only the test-designer agent need

be upgraded, rather than retraining or reprompting the entire system.

Benefits: error detection and correction. Multi-agent systems create multiple checkpoints where errors can be caught. A reviewer agent can identify code smells or logic bugs before execution. A separate test executor can detect runtime failures and provide diagnostic feedback to the programmer without the programmer needing to self-critique. This separation mirrors human code review and pair programming, which are empirically known to improve code quality. MapCoder’s debug agent, for example, systematically analyses test failures and proposes fixes, iterating until the code passes. Single-agent systems can approximate this through self-reflection [15], but multi-agent designs make the separation explicit and avoid the risk of self-confirming biases.

Costs: coordination and token overhead. The main cost is coordination complexity. Each inter-agent communication requires prompt construction, LLM inference, and state synchronisation. If agents operate sequentially, total latency is the sum of all agent runtimes. If agents operate in parallel (e.g., multiple programmers proposing solutions), orchestration logic must merge or vote on outputs. Token costs scale with the number of agents and communication rounds: MetaGPT and ChatDev can consume several times the tokens of a single-agent baseline. In practice, frameworks mitigate this by caching shared context, compressing intermediate artifacts, or limiting the number of iterations. However, for simple tasks where a single-agent solution suffices, the overhead is not justified.

Failure modes: handoff errors and misalignment. Multi-agent systems introduce new failure modes. Handoff errors occur when one agent’s output is misunderstood or ignored by the next. For example, if the architect agent specifies an API contract in natural language and the programmer misinterprets it, the resulting code may be internally consistent but wrong. MetaGPT’s structured artifacts reduce this risk by enforcing explicit schemas, but free-form chat-based systems like ChatDev remain vulnerable. Another failure mode is *agent misalignment*: if the test designer agent generates trivial or incorrect tests, the system may converge on code that passes those weak tests but fails in production. AgentCoder addresses this by ensuring tests are written independently, but the quality still depends on the test-designer prompt and model.

Diminishing returns with agent count. Adding more agents does not monotonically improve performance. Beyond a certain point, coordination overhead dominates and additional roles contribute little. Empirical studies on ChatDev and MetaGPT suggest that 3–5 agents (planning, coding, testing, review) capture most of the benefit, while systems with 10+ agents show marginal improvements at much higher cost. The optimal agent count depends on task complexity: simple function-writing benefits from programmer-tester pairs, while full software projects may justify architect-programmer-tester-reviewer-documenter chains. However, even for complex tasks, the evidence does not yet support very large agent counts analogous to human teams of dozens.

When to choose multi-agent designs. Multi-agent frameworks are most beneficial when: (i) the task involves multiple distinct phases (design, implementation, validation) that benefit from specialisation; (ii) intermediate artifacts (specifications, test suites) can be produced and validated independently; (iii) the base LLM is strong enough that role-specific prompting yields measurably different behaviours; and (iv) the cost overhead is acceptable given the quality improvement. For simple one-off code snippets, single-agent systems with self-refinement suffice and are faster. For end-to-end software engineering tasks on benchmarks like SWE-bench, the evidence is mixed: some multi-agent systems underperform well-tuned single-agent scaffolds (recall Agentless outperforming many agentic designs in §3), suggesting that scaffold quality matters as much as agent count. The frontier remains open, and multi-agent research con-

tinues to refine orchestration protocols and role definitions.

6 Execution Feedback and Iterative Refinement

Code generation rarely succeeds in one shot. Even human programmers iterate: write, compile, test, debug, revise. Agentic systems close this loop by executing generated code, observing runtime outcomes—test failures, compiler errors, exceptions—and revising their output accordingly. This section examines the feedback signals agents consume (§6.1), the mechanisms by which they refine code (§6.2), the techniques for localising and repairing faults (§6.3), and the criteria for deciding when to stop iterating (§6.4).

6.1 Feedback Signals and Environments

Agents require feedback to improve. The type, source, and presentation of feedback critically affect refinement efficacy.

Types of feedback. The most common feedback is *test execution results*: pass/fail verdicts, failing test cases, assertion messages, and expected-versus-actual output diffs. When tests fail, agents receive concrete evidence of incorrectness. A richer signal is *compiler or interpreter errors*: syntax errors, type mismatches, undefined-variable messages, and stack traces. These pinpoint the location and nature of faults without requiring test design. Static analysis tools provide another layer: linters flag style violations, type checkers identify inconsistencies, and data-flow analysers detect null dereferences or race conditions before execution. Some systems incorporate *human feedback* in the loop: developers approve or reject patches, provide natural-language explanations of failures, or annotate bug reports. Finally, agents can generate *self-critique*: an LLM explains the generated code in natural language, identifies logical flaws, or predicts potential failures without executing the code (“rubber duck debugging”) [4].

Feedback sources: external versus internal. Feedback may come from *external* oracles—pre-written test suites, continuous-integration pipelines, runtime environments—or be *internally simulated* by the agent itself. External feedback is authoritative but requires infrastructure and may be slow or expensive. Internal feedback is fast and requires no setup, but

risks hallucination: the agent may incorrectly believe its code is correct or fabricate error messages. Hybrid approaches use external execution for validation and internal self-critique for diagnosis.

Sandboxed execution environments. To safely run untrusted generated code, systems employ sandboxing. Common techniques include *containerisation* (Docker), *virtual machines*, *language-level isolation* (restricted Python interpreters), and *timeout and resource limits*. SWE-agent and AutoCodeRover run generated code in Docker containers with filesystem restrictions and network isolation. Reflexion [19] uses a Python interpreter with timeout guards and exception handling. Sandboxing prevents malicious code from damaging the host system, but introduces latency and limits access to system resources, databases, or network services that the real code may need. Balancing safety and realism is an ongoing design challenge.

6.2 Iterative Refinement Mechanisms

Once feedback is available, agents must incorporate it to produce better code. Refinement strategies vary in how they represent feedback, how they prompt the model to revise, and whether they maintain memory across iterations.

Reflexion: verbal reinforcement learning. Reflexion [19], from Northeastern, MIT, and Princeton, treats refinement as a form of reinforcement learning without weight updates. After each trial, the agent reflects on what went wrong by generating a natural-language critique (“The function failed because it did not handle negative inputs”). This reflection is stored in an episodic memory buffer and retrieved in subsequent attempts, conditioning the model to avoid past mistakes. Reflexion accepts diverse feedback: scalar rewards (e.g., pass rate), binary verdicts, or free-form error messages. The key innovation is converting feedback into *verbal* form that persists across trials, enabling the model to learn from experience within a single episode. On HumanEval, Reflexion achieves 91% pass@1, surpassing single-shot GPT-4 (80%) by explicitly encoding trial-and-error history. The approach generalises beyond code: it improves sequential decision-making in games and language reasoning tasks.

Self-debugging: rubber duck debugging. Chen et al. [4] propose *self-debugging*, where the

model critiques and repairs its own code without external feedback. The agent generates code, then *explains* the code line-by-line in natural language. This explanation serves as a self-generated specification, allowing the model to spot discrepancies between intent and implementation. If the explanation reveals a flaw (“This loop iterates over indices, but I wanted to iterate over values”), the model revises the code. Self-debugging also incorporates external execution feedback when available: the model receives error messages or failing test outputs and reasons about root causes before proposing fixes. On Spider (text-to-SQL), self-debugging improves baseline accuracy by 2–3% without unit tests and by up to 12% on MBPP when tests are available. The method demonstrates that internal self-critique complements external execution signals.

Repair prompts and diff-based editing. Simpler refinement strategies use *repair prompts*: the agent receives the original task, the buggy code, and the error message, then generates a revised version. To reduce token costs, some systems use *diff-based editing*, where the model outputs only the changes (insertions, deletions) rather than the full code. This is particularly effective for large files: instead of regenerating hundreds of lines, the agent specifies “replace line 42” or “insert after line 57”. SWE-agent’s edit tool and Agentless’s diff-based repair both adopt this paradigm. The trade-off is that models must learn the diff format, and parsing errors can break the repair loop.

Multi-turn conversation with the environment. Some agents treat refinement as a dialogue with the execution environment. The agent generates code, the environment returns feedback, the agent asks clarifying questions or requests additional context (“What is the expected output for negative inputs?”), the environment answers, and the cycle repeats. This conversational pattern mirrors human debugging sessions with a collaborator. However, it assumes the environment can answer semantic questions, which is typically not the case for automated test suites. In practice, multi-turn systems interact with a supervisor LLM that interprets test failures and generates explanations, adding an extra layer of inference cost.

6.3 Fault Localization and Self-Repair

When refinement targets bugs in existing code rather than improving draft solutions, it becomes *automated*

program repair (APR). The classic APR pipeline has three stages: fault localisation (identify buggy lines), patch generation (propose fixes), and patch validation (test the fix). LLM-based agents challenge this structure by generating end-to-end repairs.

Traditional fault localisation. Pre-LLM APR systems rely on *spectrum-based fault localisation* (SBFL), which ranks code lines by suspiciousness scores derived from test coverage: lines executed by failing tests and not by passing tests receive high scores. Ochiai, Tarantula, and DStar are standard SBFL metrics. LLM-based systems initially adopted this pattern, using SBFL to extract a small code snippet and prompting the model to repair it. However, fault localisation is imperfect: even oracle-provided bug locations (“perfect FL”) do not guarantee correct repairs, and SBFL scores are often noisy.

FL-free repair with objective alignment. Recent work questions whether explicit fault localisation is necessary. D4C (Direct Debug Drives Decent Code) [26] argues that LLMs should repair entire functions rather than isolated hunks, aligning the task with their pre-training objective (next-token or infilling completion). D4C prompts the model with the full function, failing test cases, and error messages, and asks it to output a corrected function. On Defects4J, D4C repairs 180 single-function bugs with only 10 samples per bug, outperforming systems with perfect fault localisation by 10% and reducing sample count by 90%. The insight is that LLMs, trained on vast corpora of correct code, implicitly learn to locate faults when given sufficient context; forcing them to target specific lines may constrain their reasoning.

Stack-trace analysis and root-cause reasoning. When execution fails, agents receive stack traces showing the call chain and exception type. Stack traces are rich signals: they identify the failing function, reveal the exception (NullPointerException, IndexError), and often include variable values. Agents can parse stack traces to infer root causes: “IndexError at line 15 suggests the list is shorter than expected; check loop bounds.” SelfEvolve [12] and AgentCoder iteratively refine code by feeding error messages back to the LLM, which reasons about the fault and proposes targeted edits. The challenge is that stack traces can be noisy (multiple exceptions, deep call stacks) and may not reveal semantic bugs (e.g., off-by-one errors that do not crash).

Test-driven repair loops. A refinement loop driven by test feedback proceeds as follows: (1) generate initial code, (2) run tests, (3) if any fail, extract failure messages, (4) prompt the model to fix the code, (5) repeat until all tests pass or a budget is exhausted. The stopping condition is critical: systems may iterate 1, 3, 5, or 10 times. Empirically, most gains occur in the first 2–3 iterations; further iterations yield diminishing returns and risk introducing new bugs. MapCoder allocates separate iterations for debugging, while MetaGPT’s feedback module runs until tests pass or a maximum round is reached. The iteration budget balances quality and cost.

Candidate ranking and ensemble repair. When multiple repair attempts produce different patches, agents must select the best one. Ranking strategies include: (i) test coverage (choose the patch that passes the most tests), (ii) LLM-as-judge (ask the model to explain and rank patches by confidence), (iii) ensemble voting (generate N patches and pick the most common or most syntactically similar to the original), and (iv) semantic similarity (prefer patches that minimally alter the original code). AlphaRepair and CodeT use candidate ranking with test execution to filter implausible patches before final selection.

6.4 Convergence and Stopping Criteria

Iterative refinement must terminate. Agents need stopping criteria that balance solution quality, cost, and the risk of overfitting to weak test suites.

Pass-rate thresholds. The simplest criterion is *full test pass*: stop when all provided tests pass. This works well for competitive programming (HumanEval, CodeContests) where test suites are comprehensive. However, in software engineering tasks (SWE-bench), tests may be incomplete or incorrect. A patch that passes all tests may still be wrong (a “plausible but incorrect” patch), and a correct patch may fail tests due to environment issues. Some systems use a *relaxed threshold*: stop when $\geq 80\%$ of tests pass, trading coverage for faster convergence.

Diminishing improvement detection. If successive iterations yield no improvement—measured by test pass rate, execution success, or LLM confidence scores—the agent should stop. For example, if iterations 3, 4, and 5 all produce code that fails the same test with the same error, further iteration is unlikely

to help. SelfEvolve monitors improvement deltas: if the pass rate increases by less than 1% across two iterations, refinement halts. This heuristic prevents endless loops where the model oscillates between two incorrect states.

Budget limits: iterations and tokens. Practical systems impose hard limits. Iteration budgets (3, 5, 10 rounds) cap the number of refinement cycles. Token budgets limit total inference cost: after consuming a threshold (e.g., 50k tokens), the agent returns its best attempt. Reflexion and Agentless both use iteration caps, while D4C’s low sample count (10 tries per bug) implicitly limits token use. The optimal budget depends on task difficulty: simple bugs may resolve in one iteration, while complex logical errors may require many.

Risk of overfitting to weak test suites. A critical failure mode is overfitting. If the test suite is weak (few tests, shallow coverage), the agent may generate code that narrowly passes those tests but fails on unseen inputs. This is the “plausible patch” problem in APR: patches that satisfy the validation oracle but do not generalise. Mitigations include: (i) *test augmentation*, where the agent or a separate test generator produces additional test cases to catch edge cases; (ii) *static checks*, requiring generated code to satisfy linting, type checking, or invariant assertions beyond test pass; (iii) *semantic similarity constraints*, penalising patches that radically alter program structure; and (iv) *LLM-based verification*, asking the model to critique the patch or predict whether it will generalise. However, no technique fully eliminates overfitting, and the problem remains open.

Human-in-the-loop approval. In high-stakes domains (medical software, financial systems), automated iteration may be unacceptable without human oversight. Systems like OpenHands and Devin allow developers to inspect intermediate outputs, approve or reject patches, and provide guidance before the next iteration. This trades speed for safety and introduces latency, but ensures that convergence criteria align with human judgment rather than proxy metrics.

7 Discussion

7.1 Consolidated Patterns and Trends

Recurring design patterns across all five topics (observe-think-act loops, tool orchestration, LLM-as-judge verification). Macro trends: from single-shot prompting to agentic scaffolds, from monolithic agents to composable pipelines, from flat CoT to tree-structured search.

7.2 The Evaluation Landscape

Cross-cutting view of benchmarks (HumanEval, MBPP, SWE-bench, Defects4J, CrossCodeEval), their coverage of pipeline stages, saturation concerns, and gaps in realism.

7.3 The Context Window Bottleneck

How finite context constrains every pipeline stage; retrieval-augmented generation, hierarchical summarization, and repository-level chunking as mitigations.

7.4 Open Challenges

Unsolved problems: long-horizon reasoning, specification ambiguity, cost efficiency, trustworthiness, and human-in-the-loop control design.

7.5 Implications for Software Engineering Practice

How these advances may reshape developer workflows, team structures, and the economics of software production.

8 Conclusion

Summarize key findings across the five research fronts, restate the pipeline framing as a unifying lens, and point to the most promising future directions.

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